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The Editors

Superconductor Science and Technology

Submission of a paper: “Damage–repair kinetics and phase stiffness in a site-diluted XY model: implications for irradiated superconductors”

Dear Editors,

I submit the enclosed paper for consideration in Superconductor Science and Technology. It asks what a minimal model can say about the competition between displacement damage and recovery in irradiated superconductors, the subject of recent work published in this journal on REBCO conductors for fusion magnets [Prokopec et al 2015, Fischer et al 2018, Unterrainer et al 2022, Adams et al 2023]. In those studies the transition temperature falls linearly with displacement damage, part of it is recovered by annealing, and the degradation is attributed to a loss of superfluid density. The paper takes the phase stiffness, the counterpart of the superfluid density, as its observable.

Main results

- A mean-field reduction of damage–repair kinetics, re-derived from the implemented update for the damaged fraction, predicts bistability. On the lattice, with up to 10 seeds per condition and sizes to $L = 64$, no hysteresis and no dependence on the initial state is resolved (largest difference 0.0027 in the coherence, none significant after Holm correction; quasi-static ramps close), because the closure of the reduction fails: the multiplier of the damage rate that the lattice applies is a smooth function of the damaged fraction. The statement is made for the ranges tested.
- Without a field, and with repair that preserves the phase symmetry, repair restores spontaneous phase coherence and a stiffness, measured as the twist response of the non-equilibrium steady state with the winding number recorded, above a threshold ratio of repair to damage; finite-size scans (sizes to 96 and 64, operational cutoffs, not a thermodynamic-limit determination) place it at $\rho = 7.00\text{--}7.75$ (repair that leaves the phase unchanged) and $5.50\text{--}6.00$ (repair to the phase of the neighbors), against about 2.5 without state-dependent damage. The equilibrium helicity-modulus formula underestimates this stiffness by up to about 30% and is negative where the stiffness vanishes. The twist response is validated against the equilibrium helicity modulus on clean and diluted lattices, with imposed windings, and against an independent static code.
- Numerically, the repair-to-damage ratio controls the state ever better as damage becomes slow compared with phase relaxation: three repair rules that differ by 0.12 in the coherence at finite rates agree within 0.012 at the slowest damage scanned (the limit itself is not claimed), and the equilibrium formula approaches the twist response.
- Static random dilution of the lattice loses stiffness at a removed fraction of 0.25 ± 0.01 (sizes to 96, finite-size extrapolation and its sensitivity reported); compact damage clusters tolerate considerably more. An illustrative effective mapping to displacement damage gives about 10 model sites per dpa (a

conversion factor, with no further physical interpretation), and an illustrative annealing analysis is compatible with a broad activation spectrum, whose width the two reported recoveries do not determine.

The model is phenomenological and is not calibrated to any conductor; it has no pinning, so it says nothing about the critical current or its initial rise with fluence, and I have said so in the paper. The experimental values quoted are taken from the cited papers. The paper closes with a proposed fission-reactor irradiation test (the same accumulated damage reached with different damage-rate histories, and different defect morphologies) that could falsify its kinetic claims.

Data, code and disclosures

All code, raw data and analysis scripts are public at <https://github.com/sandlerleon/damage-repair-xy-stiffness>, archived at <https://doi.org/10.5281/zenodo.23173811>; the manuscript is available as a preprint at <https://doi.org/10.5281/zenodo.23173812>. The simulation engine reproduces the update rules of the open-source MyUncle framework bit for bit. This manuscript has not been published and is not under consideration by any other journal. I declare no competing interests and no funding, and the use of AI-assisted tools is declared in the manuscript.

Thank you for considering the paper.

Yours sincerely,

Leon Sandler